

Miles Levine
Homework 5 documentation

Question 1 - zodiac chi-square goodness of fit test

Null hypothesis: **The distribution of zodiac signs among the heads of large companies is consistent with what would be expected by chance.**

Alternative hypothesis: **The distribution of zodiac signs among the heads of large companies is not equally divided; some zodiac signs are overrepresented or under-represented.**

Result: **Fail to reject the null hypothesis. The P-value was greater than the significance level for the Chi-square test. The birthdays of heads of fortune 500 companies are equally distributed and divided.**

```
> chisq.test(zodiac$Births)

      Chi-squared test for given probabilities

data:  zodiac$Births
X-squared = 5.0938, df = 11, p-value = 0.9265
```

Question 2- analgesics one-way ANOVA

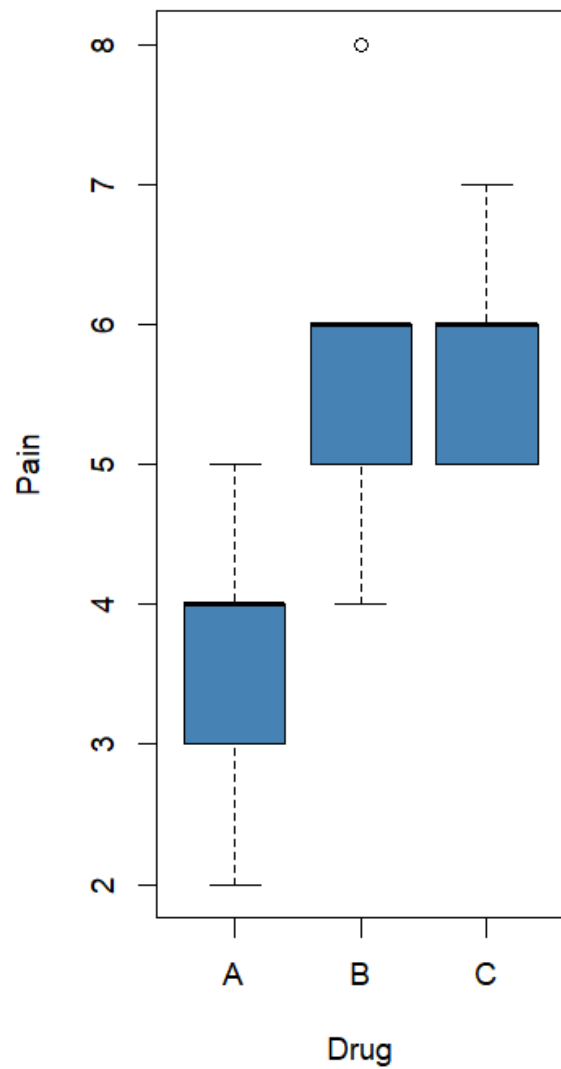
omnibus null: **There is no significant difference in mean pain relief effectiveness among the three drug formulations for migraine headache sufferers.**

All of the drug formations have the same effectiveness on migraine headache sufferers

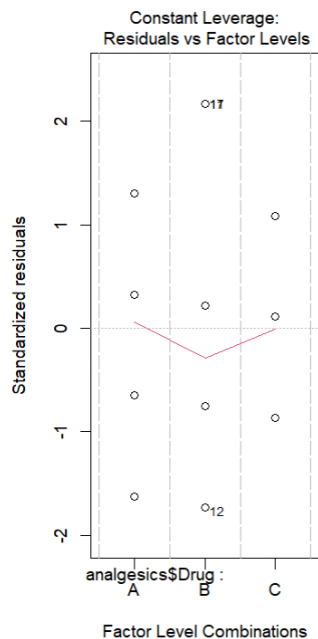
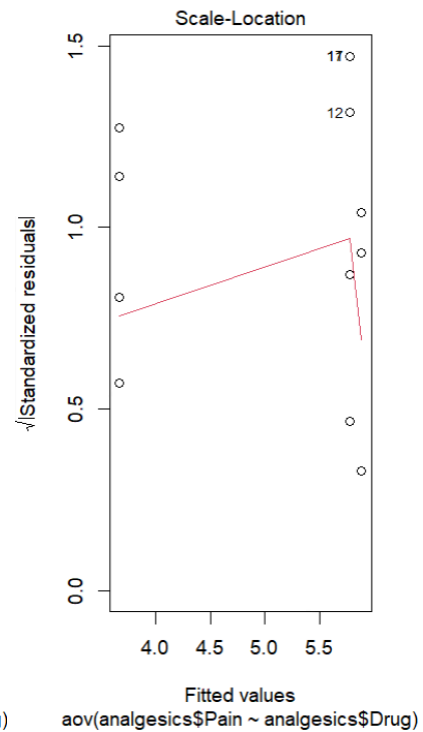
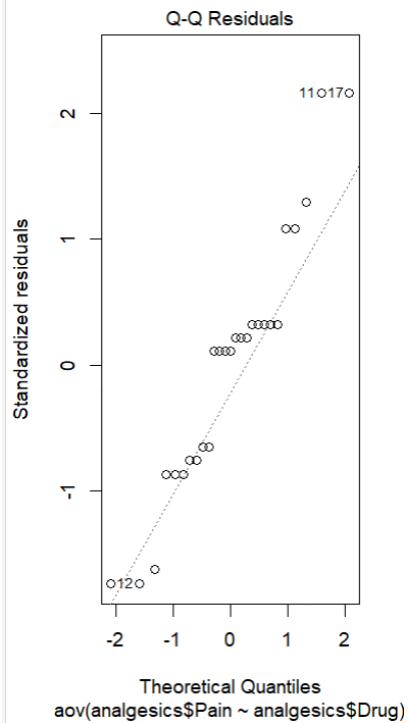
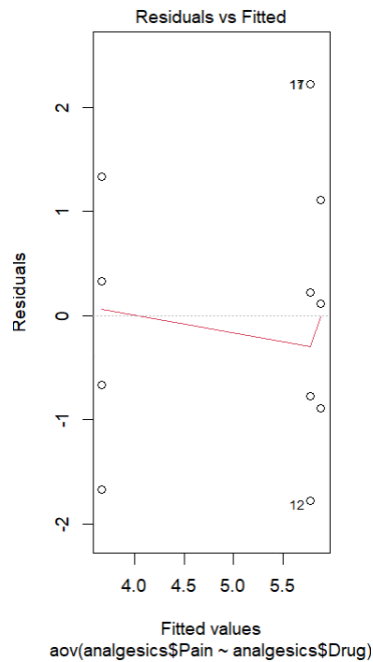
Alternative: **There is a significant difference in mean pain relief effectiveness among at least two of the three drug formulations for migraine headache sufferers.**

At least one of the 3 drug formulations has different headache pain level effects

Pain level by Analgesic Drug



The assumptions do not look to be satisfied since the boxplot shows an outlier and the boxes are disfigured because of skewness.



I used the transformTukey method for transforming the pain scale data.

Results: The results from the one-way ANOVA test were that the p-value is 0.000256 so reject the null hypothesis, and conclude that at least one of the means is different.

There is a significant difference in the average pain level

Untransformed results:

```
> analgesic_1w <- aov(analgesics$Pain ~ analgesics$Drug)
> summary.aov(analgesic_1w)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
analgesics\$Drug	2	28.22	14.111	11.91	0.000256 ***
Residuals	24	28.44	1.185		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> TukeyHSD(analgesic_1w)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = analgesics$Pain ~ analgesics$Drug)

$`analgesics$Drug`
      diff      lwr      upr      p adj
B-A 2.111111  0.8295028 3.392719 0.0011107
C-A 2.222222  0.9406139 3.503831 0.0006453
C-B 0.111111 -1.1704972 1.392719 0.9745173
```

Transformed results:

```
> analgesic_2w <- aov(T_tuk ~ analgesics$Drug)
> summary.aov(analgesic_2w)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
analgesics\$Drug	2	9.736	4.868	12.4	0.000201 ***
Residuals	24	9.425	0.393		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> TukeyHSD(analgesic_2w)
  Tukey multiple comparisons of means
    95% family-wise confidence level

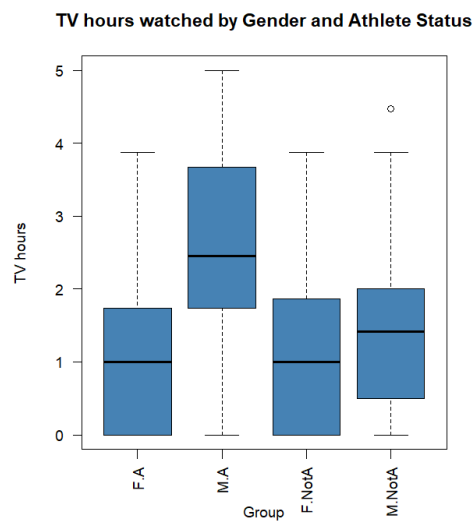
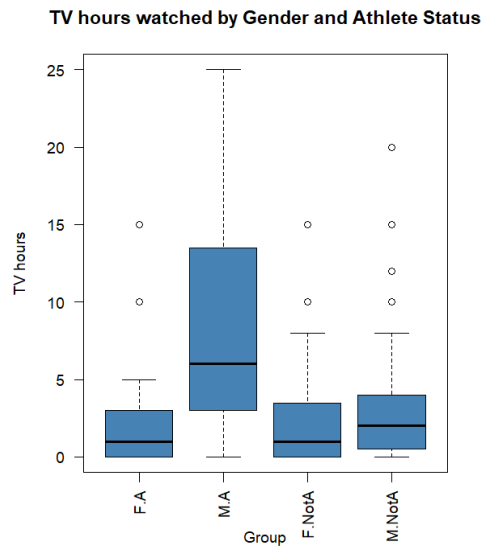
Fit: aov(formula = T_tuk ~ analgesics$Drug)

$`analgesics$Drug`
      diff      lwr      upr      p adj
B-A 1.23411109  0.4963988 1.9718234 0.0009461
C-A 1.31016490  0.5724526 2.0478772 0.0004954
C-B 0.07605381 -0.6616585 0.8137661 0.9641746
```

Question 3- tv watching

Null: There is no significant difference in the mean TV viewing hours among the groups defined by gender (male vs. female) and athletic status (athlete vs. non-athlete). In other words, There is no interaction effect between gender and athletic status on TV viewing hours.

Alternative: The alternative hypothesis is that there is a significant difference in TV-watching habits between the groups based on gender and athletic status.



Transformed:

Additive two-way ANOVA results:

```
> ##Additive
> tv_add <- aov(tv$Sq.Rt.TV ~ tv$Gender + tv$Athlete)
> summary(tv_add)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
tv\$Gender	1	13.93	13.935	10.75	0.001233	**
tv\$Athlete	1	19.20	19.199	14.82	0.000161	***
Residuals	194	251.37	1.296			

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

two-way ANOVA with interaction results:

```
> ##With Interaction
> tv_interact <- aov(tv$Sq.Rt.TV ~ tv$Gender * tv$Athlete)
> summary(tv_interact)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
tv\$Gender	1	13.93	13.935	11.34	0.000918	***
tv\$Athlete	1	19.20	19.199	15.62	0.000109	***
tv\$Gender:tv\$Athlete	1	14.11	14.114	11.48	0.000852	***
Residuals	193	237.26	1.229			

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> #Compare model output load library(AICcmodavg) if you haven't already
> model_set <- list(tv_add, tv_interact)
> model_names <- c("tv_add", "tv_interact")
> aictab(model_set,model_names)
```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
tv_interact	5	606.01	0.00	0.99	0.99	-297.85
tv_add	4	615.29	9.28	0.01	1.00	-303.54

Results: The best fit model is the tv_interact. Two-way ANOVA with interaction performs better than the additive two-way ANOVA because the AICc value is lower indicating better model fit.

Results from `tukeyHSD(tv_interact)`:

```
> TukeyHSD(tv_interact)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = tv$Sq.Rt.TV ~ tv$Gender * tv$Athlete)

$`tv$Gender`
      diff      lwr      upr      p adj
M-F 0.5324768 0.2205411 0.8444126 0.0009178

$`tv$Athlete`
      diff      lwr      upr      p adj
NotA-A -0.7217935 -1.082222 -0.3613653 0.0001096

$`tv$Gender:tv$Athlete`
      diff      lwr      upr      p adj
M:A-F:A    1.4485461 0.6232690 2.2738232 0.0000560
F:NotA-F:A  -0.0489430 -0.7489161 0.6510301 0.9978813
M:NotA-F:A   0.1562247 -0.5394238 0.8518733 0.9374042
F:NotA-M:A  -1.4974891 -2.1459211 -0.8490571 0.0000001
M:NotA-M:A  -1.2923214 -1.9360828 -0.6485600 0.0000030
M:NotA-F:NotA 0.2051677 -0.2673878 0.6777232 0.6744618
```

`tv$Gender` analysis:

The difference between male and female TV viewing hours is approximately 0.53 hours. The confidence interval (lower to upper) for this difference is [0.221, 0.844]. The p-value associated with this comparison is 0.0009178, indicating that the difference in TV viewing hours between males and females is statistically significant at a 95% confidence level.

`tv$Athlete` analysis:

The difference between varsity athletes and non-athletes in TV viewing hours is approximately -0.72 hours. This negative difference suggests that non-athletes watch slightly more TV hours than varsity athletes. The confidence interval (lower to upper) for this difference is [-1.082, -0.361]. The p-value associated with this comparison is 0.0001096, indicating that the difference in TV viewing hours between varsity athletes and non-athletes is statistically significant at a 95% confidence level.

There is a statistically significant difference in TV viewing hours between males and females.

There is a statistically significant difference in TV viewing hours between varsity athletes and non-athletes.

The findings of Tukey's HSD test offer compelling evidence of variations in TV-watching behaviors between groups according to gender and athletic status.